ESE 3105 Concept Primer

Heat and Mass Transfer · KUET · read this first: the subject from zero, before the past-paper solutions

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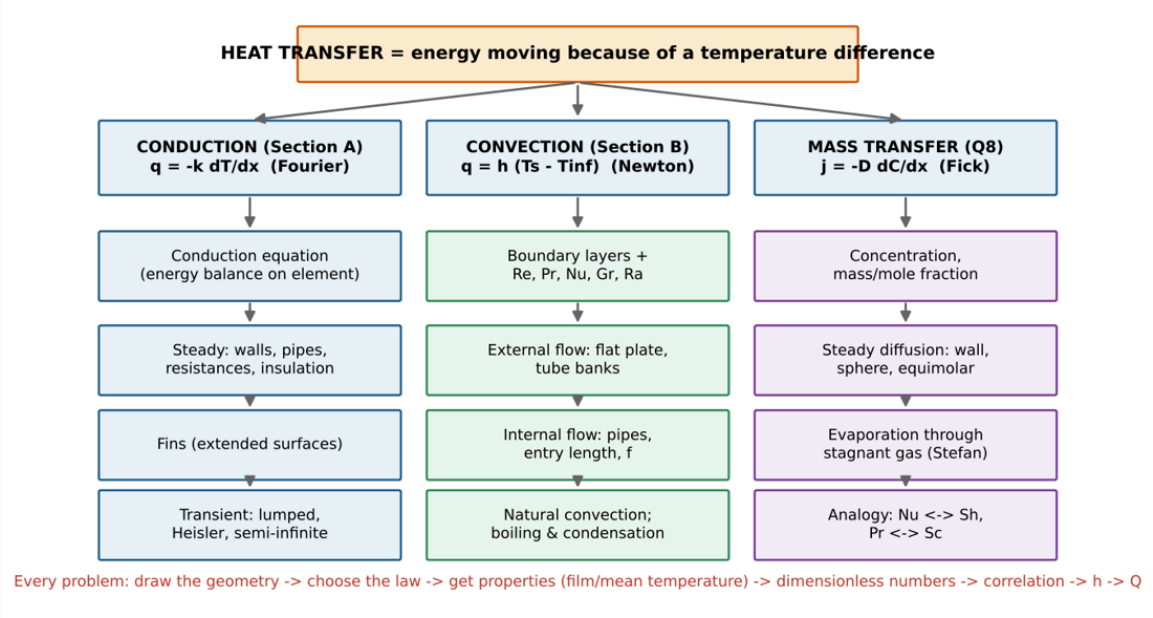
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How to use this primer

This booklet teaches heat and mass transfer from zero, in the order the exam needs it. Read it once before the Complete Solutions. Each section has a green **core idea** box (learn this first), the key terms, the formulas with every symbol and unit, and a short worked example. Section 2 is a maths and units refresher: most marks lost in this course come from a wrong property table, a wrong length (D or r, height or A/p), or °C used where K is needed.

The pass set (from the Question Map): Section A topics A1 (modes and the conduction equation), A2 (walls, pipes, resistances), A3 (critical radius), A4 (fins) and A6 (boiling and condensation); Section B topics B1 (convection basics), B3 (flat-plate numericals), B5 (tube banks), B6 (natural convection) and B7 (mass transfer). Those topics are enough to write three full questions in each section.

The whole course on one page



Concept map of ESE 3105: three modes, conduction problems, convection problems, phase change and mass transfer

Read the map from top to bottom: heat moves by **conduction** (through solids), **convection** (to a moving fluid) and **radiation**. Conduction problems are steady (walls, pipes, fins, insulation) or transient (bodies heating up or cooling). Convection problems are forced (plates, pipes, tube banks) or natural (buoyancy). Boiling and condensation are convection with phase change. Mass transfer is the same mathematics with concentration instead of temperature.

Maths, physics and units refresher

> Almost every numerical follows one pattern: **draw the picture → pick the right formula/correlation → get properties at the right temperature → substitute in SI units → check the answer is sensible**.

Units you must convert without thinking

Quantity	Conversions
Length, area	1 cm = 0.01 m, 1 mm = 0.001 m; 1 cm^2 = 10^{-4} m^2; 1 cm^3 = 10^{-6} m^3 (so 50 W/cm^3 = 5×10^7 W/m^3)
Temperature	T(K) = t(°C) + 273.15. Use K in radiation (T^4), in $\beta = 1/T_f$ and in $C = P/R_uT$. A temperature *difference* is the same in K and °C.
Power, energy	1 W = 1 J/s; 1 kW = 1000 W; 1 kJ = 1000 J; h_{fg} in kJ/kg must be ×1000 in J/kg inside correlations
Pressure	1 atm = 101.325 kPa; 1 kN/m^2 = 1 kPa
Velocity	15 m/min = 0.25 m/s
Diffusivity	1 cm^2/s = 10^{-4} m^2/s

Logs, exponentials and roots

- $\ln(a/b) = \ln a - \ln b$; e^{-x} for x small $\approx 1 - x$. Cooling laws: $\theta/\theta_i = e^{-bt}$; to find t: $t = -\ln(\theta/\theta_i)/b$.
- Quarter powers: $x^{1/4} = \sqrt{(\sqrt{x})}$. Example $10^8 \rightarrow 100$ ($10^8 = (10^2)^4$).
- Hyperbolic functions: $\sinh x = (e^x - e^{-x})/2$, $\cosh x = (e^x + e^{-x})/2$, $\tanh x = \sinh x/\cosh x$; tanh 1 = 0.762, tanh 2 = 0.964, tanh 3 = 0.995.
- Error function erf(η): read from a table; erf(0) = 0, erf(0.37) = 0.40, erf(0.5) = 0.52, erf(1) = 0.84, erf(2) = 0.995.
- Log mean temperature difference: $\Delta T_{lm} = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$; always between the two differences.

Calculus you need

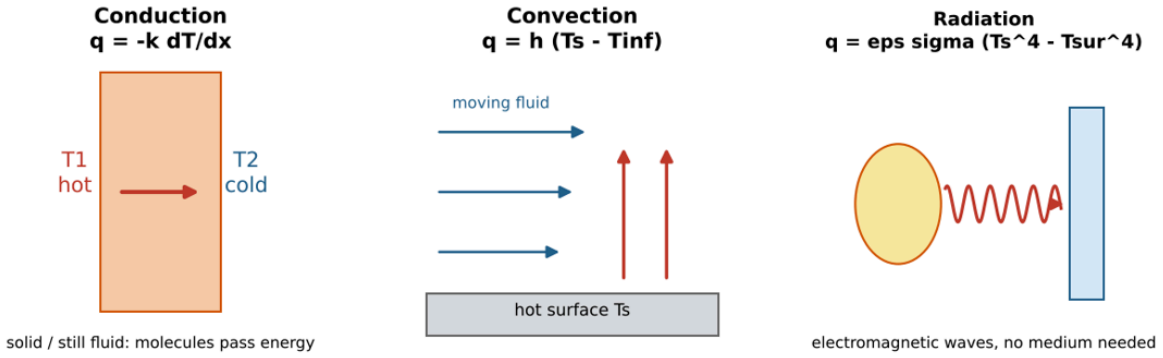
- $d/dx(x^n) = nx^{n-1}$; $\int x^n dx = x^{n+1}/(n+1)$; $\int dx/x = \ln x$.
- Equation $d^2\theta/dx^2 = m^2\theta$ has the solution $\theta = C_1e^{mx} + C_2e^{-mx}$ (fins).
- Equation $d^2T/dx^2 = 0$ gives $T = C_1x + C_2$ (plane wall); $(1/r)d/dr(r\ dT/dr) = 0$ gives $T = C_1\ln r + C_2$ (pipe).
- Average of a function over 0 to L: $\bar{f} = (1/L)\int_0^L f\ dx$. If $f \propto x^{-1/2}$, $\bar{f} = 2f(L)$.

Reading the property tables

1. Decide the temperature: **film** $T_f = (T_s + T_\infty)/2$ for external flow and natural convection; **bulk mean** $(T_i + T_e)/2$ for pipes and tube banks; liquid film at $(T_{sat} + T_s)/2$ for condensation.
2. Read k, ν (or μ and ρ), Pr, c_p from the air or water table (Cengel A-15, A-9), interpolating linearly: $y = y_1 + (x - x_1)(y_2 - y_1)/(x_2 - x_1)$.
3. Gas at a pressure other than 1 atm: k, μ, c_p, Pr unchanged; $\rho \propto P$, so $\nu \propto 1/P$.
4. For saturated water (boiling, condensation) use the saturation table: T_{sat}, h_{fg}, ρ_v.

Conduction and the conduction equation (A1)

> Conduction is heat passing through a material from hot to cold by molecular collisions and free electrons. **Fourier's law** $Q = -kA\ dT/dx$: the heat rate is proportional to the area and the temperature gradient. The **conduction equation** is energy conservation for a tiny element: heat in − heat out + heat generated = heat stored.



Conduction, convection and radiation

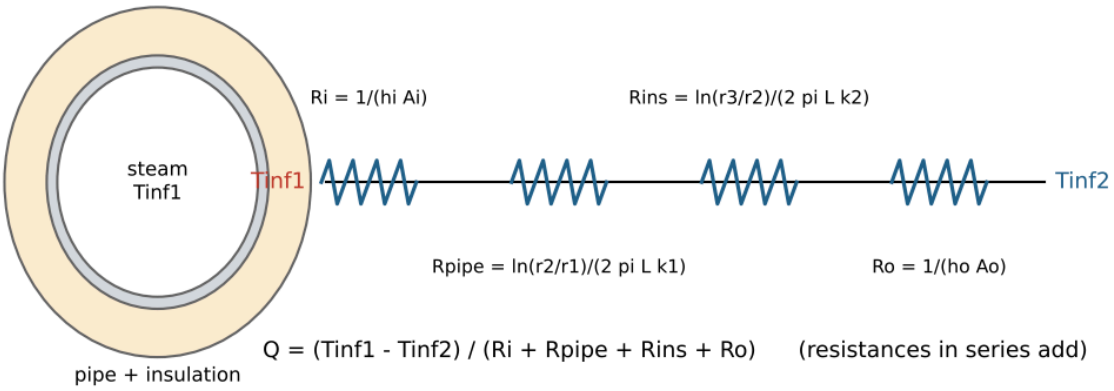
$$\partial^2T/\partial x^2 + \partial^2T/\partial y^2 + \partial^2T/\partial z^2 + \dot{e}_{gen}/k = (1/\alpha)\partial T/\partial t$$

Symbol	Meaning	Unit
Q	heat transfer rate	W
q	heat flux Q/A	W/m^2
k	thermal conductivity	W/m K
A	area normal to heat flow	m^2
α	thermal diffusivity $k/\rho c_p$	m^2/s
$\dot{e}_{gen}$	heat generated per unit volume	W/m^3
ρ, c_p	density, specific heat	kg/m^3, J/kg K
t	time	s

Example: a 0.2 m wall (k = 1.2) with faces at 120 and 50 °C, area 15 m^2: $Q = 1.2 \times 15 \times 70/0.2 = 6300$ W.

Thermal resistance networks (A2)

> For steady one-dimensional heat flow, every layer behaves like an electrical resistor: $Q = \Delta T/R$. Resistances in series add. Convection at a surface is also a resistance $1/(hA)$. So a wall with air on both sides is three resistors in a row.



Resistance network of an insulated pipe

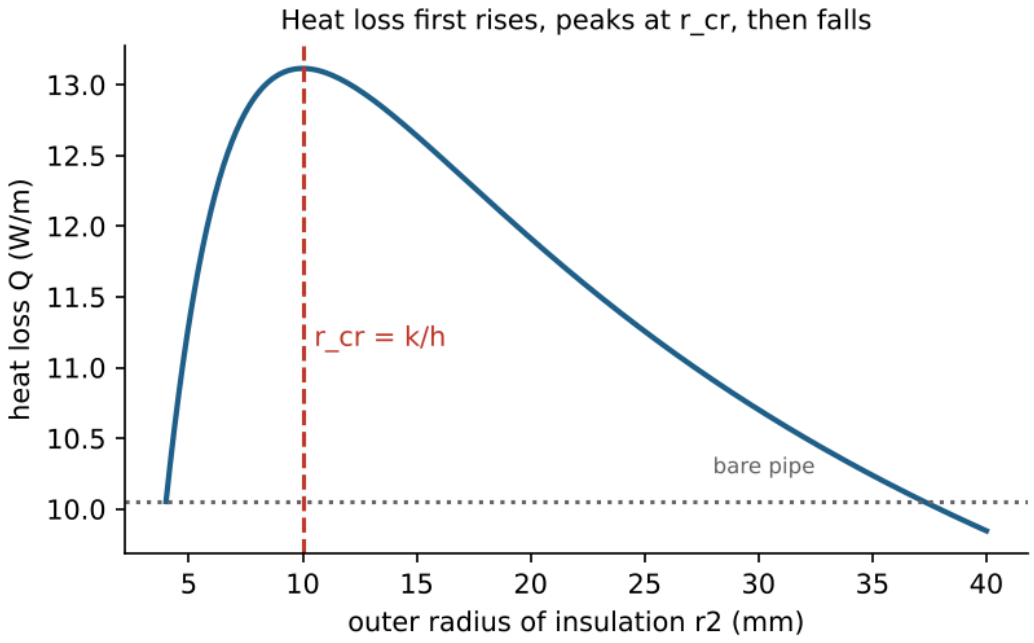
Element	Resistance R (K/W)
Plane wall	$L/(kA)$
Convection film	$1/(hA)$
Cylindrical layer	$\ln(r_2/r_1)/(2\pi kL)$
Spherical shell	$(r_2 - r_1)/(4\pi k r_1 r_2)$
Overall	$UA = 1/\Sigma R$

Symbol	Meaning	Unit
R	thermal resistance	K/W
h	heat transfer coefficient	W/m^2K
U	overall heat transfer coefficient	W/m^2K
r_1, r_2	inner, outer radius	m
L	thickness (wall) or length (pipe)	m

Example: steam pipe with glass wool: R_{inside} 0.106 + R_{pipe} 0.0002 + $R_{insulation}$ 2.348 + $R_{outside}$ 0.154 = 2.608 K/W; Q = 315/2.608 = 121 W per metre. The insulation does almost all the work.

Critical radius of insulation (A3)

Adding insulation to a thin pipe adds conduction resistance but also enlarges the outer surface, which lowers the convection resistance. Total resistance is lowest (heat loss highest) at the **critical radius** r_{cr} = k/h for a cylinder and 2k/h for a sphere.

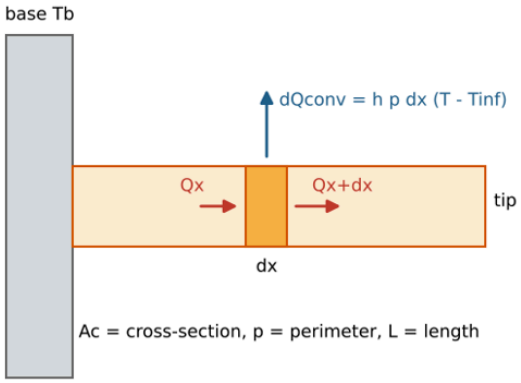


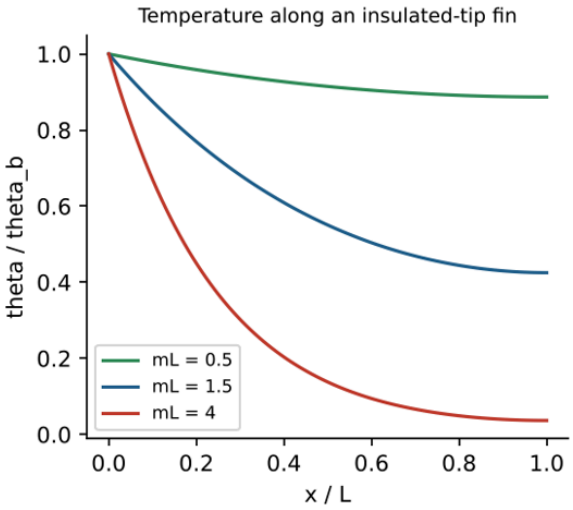
Heat loss vs outer radius of insulation

Example: k = 0.05 W/m K, h = 5 W/m^2K: r_{cr} = 0.01 m = 1 cm. A 5 mm wire covered up to 1 cm loses *more* heat than a bare one.

Fins (A4)

A fin is extra surface that conducts heat out from a wall and convects it to the air. Its temperature falls along its length, governed by $d^2\theta/dx^2 = m^2\theta$ where $\theta = T - T_\infty$ and $m = \sqrt{(hp/kA_c)}$. The larger mL, the more the tip cools down and the less it helps.





Fin element and temperature fall along the fin

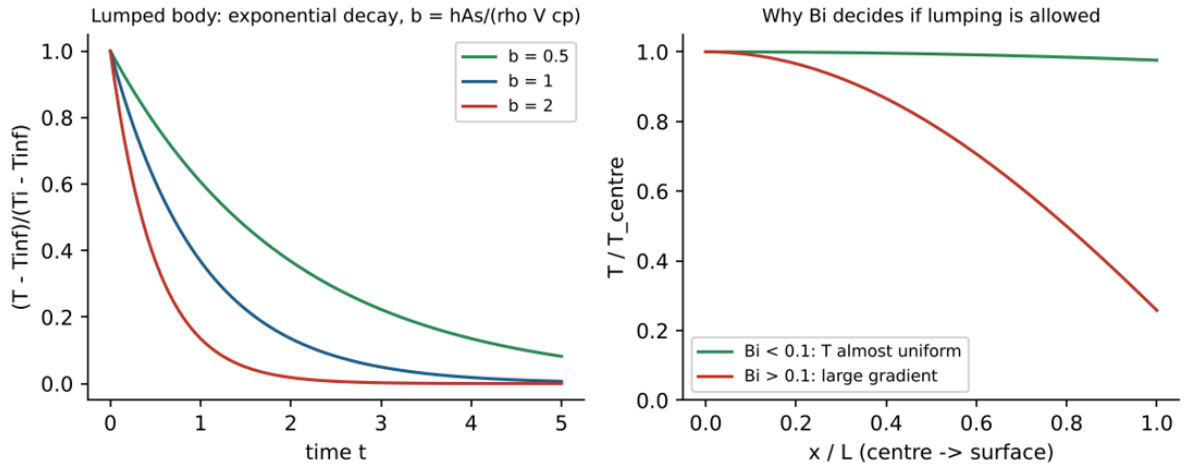
$Q_{fin} = \sqrt{(hpkA_c)}\ \theta_b \tanh(mL),\ n_{fin} = \tanh(mL)/(mL)$

Symbol	Meaning	Unit
p	perimeter of the fin cross-section	m
A_c	cross-sectional area	m^2
m	fin parameter $\sqrt{(hp/kA_c)}$	1/m
θ_b	$T_{base} - T_\infty$	K
n	efficiency Q/Q_{max}	–
ε	effectiveness $Q_{fin}/Q_{no\ fin}$	–

Example: steel rod D = 2 cm, k = 50, h = 64: $m = \sqrt{(4 \times 64/(50 \times 0.02))} = 16\ m^{-1}$; L = 0.25 m gives mL = 4, tanh 4 ≈ 1: it behaves as an infinitely long fin, Q = 25.1 W for θ_b = 100 °C.

Transient conduction (A5)

When a body is suddenly put in a hotter or colder fluid, its temperature changes with time. If its inside conducts much better than the surface convects (**$Bi = hL_c/k < 0.1$**), the whole body has one temperature that approaches the fluid temperature exponentially. Otherwise use the one-term (Heisler) solution or, for a very thick body, the semi-infinite erf solution.



Lumped cooling

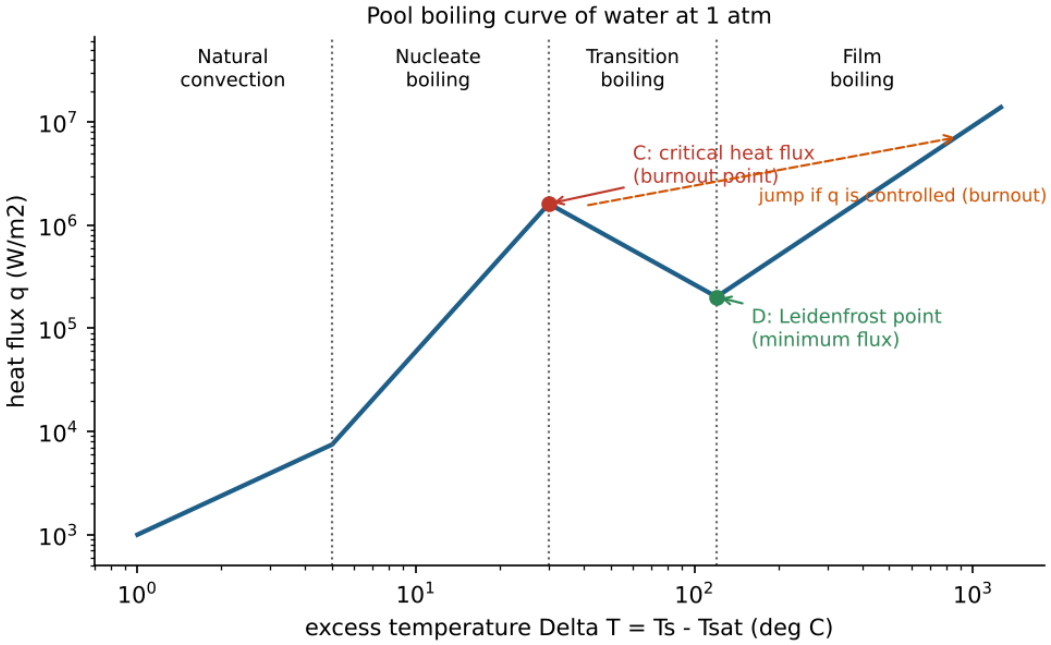
$(T - T_\infty)/(T_i - T_\infty) = e^{-bt}$, $b = hA_s/(\rho V c_p)$

Symbol	Meaning	Unit
Bi	Biot number hL_c/k (inside resistance / surface resistance)	–
Fo, τ	Fourier number $\alpha t/L_c^2$ (dimensionless time)	–
L_c	characteristic length V/A_s	m
b	inverse time constant	1/s
λ_1, A_1	one-term coefficients from tables	–
erf	error function	–

Example: a 12 cm copper ball (k = 401) in air with h = 15: $L_c = D/6 = 0.02$ m, $Bi = 15 \times 0.02/401 = 0.00075$, so lumped analysis is excellent.

Boiling and condensation (A6)

Phase change absorbs or releases latent heat h_{fg}, so h is very large. In **pool boiling** the heat flux depends on the excess temperature $\Delta T_e = T_s - T_{sat}$ along the boiling curve (natural, nucleate, transition, film). In **condensation** vapour forms a liquid film (film-wise) or drops (drop-wise) on a cold surface; the film is a resistance.



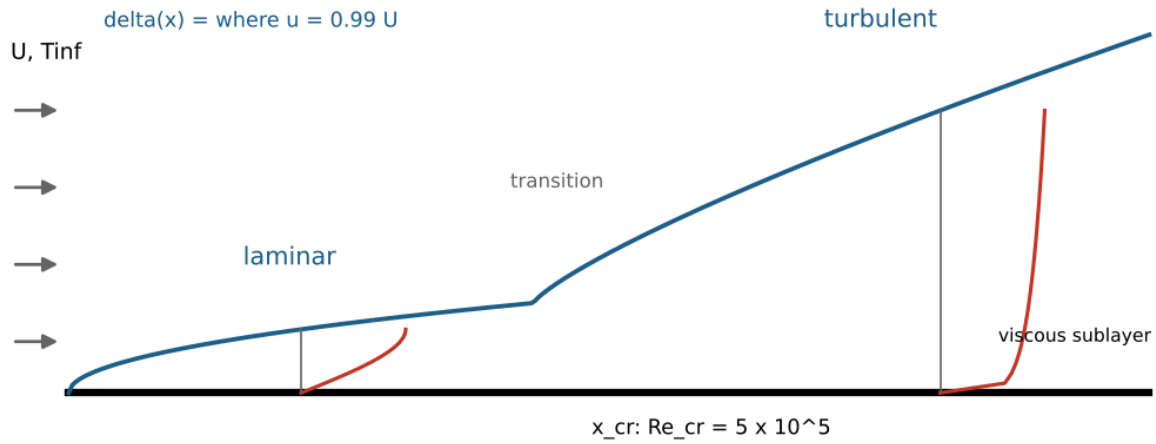
Boiling curve of water

Symbol	Meaning	Unit
T_{sat}	saturation temperature at the given pressure	°C
h_{fg}	latent heat of vaporisation	J/kg
h^*_{fg}	$h_{fg} + 0.68 c_p (T_{sat} - T_s)$, corrected for liquid subcooling	J/kg
ρ_l, ρ_v	liquid, vapour density	kg/m^3
μ_l, k_l	liquid viscosity, conductivity	kg/m s, W/m K
N	tubes in a vertical column	–
CHF (critical heat flux)	critical heat flux (peak of the curve)	W/m^2

Example: condensation rate = Q/h^*_{fg}. If a tube bank removes 163.8 kW per m and h^*_{fg} = 2468 kJ/kg, it condenses 0.0664 kg/s of steam per metre.

Convection and boundary layers (B1, B2)

At the wall the fluid is at rest and heat enters it by conduction; the flow then carries it away. Near the wall two thin layers form: the **velocity boundary layer** (where the speed rises from 0 to U_∞) and the **thermal boundary layer** (where the temperature changes from T_s to T_∞). $h \approx k/\delta_t$: thin layer = big h.



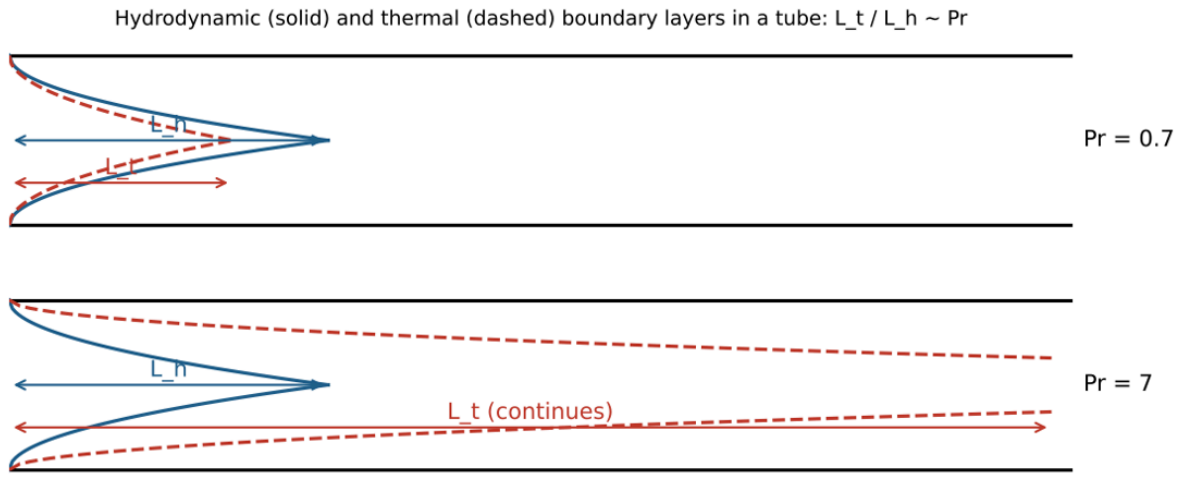
Boundary layers on a flat plate

Symbol	Meaning	Unit
Re	Reynolds number VL/ν (inertia/viscous)	–
Pr	Prandtl number ν/α	–
Nu	Nusselt number hL/k (dimensionless h)	–
ν	kinematic viscosity μ/ρ	m^2/s
δ, δ_t	velocity, thermal boundary-layer thickness	m
C_f	friction coefficient $\tau_s/(\rho V^2/2)$	–
T_f	film temperature $(T_s + T_\infty)/2$	°C

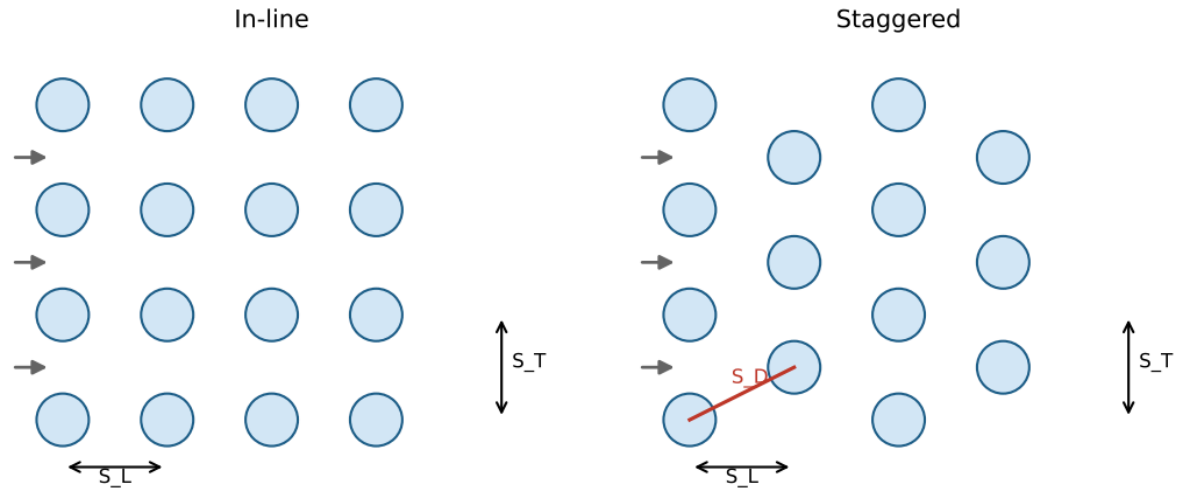
Example: air at 3 m/s over 1.2 m: $Re = 3 \times 1.2/1.896\times10^{-5} = 1.9\times10^5$ (laminar), $Nu = 0.664\sqrt{Re}\, Pr^{1/3} = 259$, $h = Nu\, k/L = 6.1\ W/m^2K$.

Forced convection: plates, pipes and tube banks (B3, B4, B5)

For each geometry there is a correlation $Nu = C\, Re^m Pr^n$. Flat plate: laminar below $Re = 5\times10^5$. Pipe: laminar below 2300, use the bulk temperature and an exit-temperature law. Tube bank: use the maximum velocity between tubes and the Zukauskas correlation.



Entry region and fully developed region in a pipe



In-line and staggered tube banks

$$T_e = T_s - (T_s - T_i)\exp(-hA_s/\dot{m}c_p),\quad Q = hA_s\Delta T_{lm}$$

Symbol	Meaning	Unit
D_h	hydraulic diameter $4A_c/p$	m
T_m	mean (bulk) fluid temperature	°C
f	Darcy friction factor	–
S_T, S_L, S_D	transverse, longitudinal, diagonal pitch	m
V_{max}	velocity in the narrowest gap	m/s
N_L, N_T	rows, tubes per row	–
ΔT_{lm}	log mean temperature difference	K

Example: tubes 1.5 cm at S_T = 5 cm, air 4.5 m/s: $V_{max} = 5 \times 4.5/(5 - 1.5) = 6.43$ m/s.

Natural convection (B6)

No fan: the fluid moves because heated fluid is lighter. The strength of this flow is measured by the Rayleigh number $Ra = g\beta\Delta T L_c^3 Pr/v^2$, which takes the place of Re. For air $\beta = 1/T_f$ in kelvin.

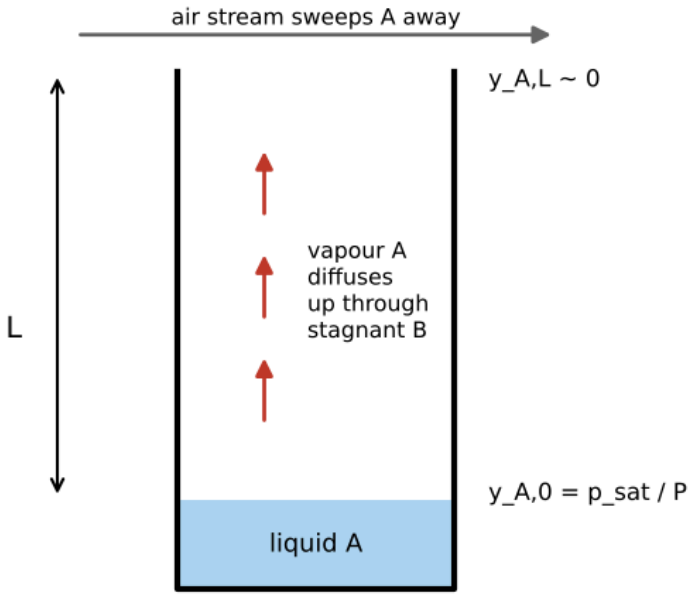
Geometry	L_c	Nu
Vertical plate	height	$0.59Ra^{1/4}$ (10^4-10^9)
Horizontal, hot face up	A/p	$0.54Ra^{1/4}$ (10^4-10^7), $0.15Ra^{1/3}$ (10^7-10^{11})
Horizontal, hot face down	A/p	$0.27Ra^{1/4}$
Horizontal cylinder	D	Churchill-Chu $\{\{0.6 + 0.387Ra^{1/6}/[1 + (0.559/Pr)^{9/16}]^{8/27}\}\}^2$

Symbol	Meaning	Unit
g	gravity 9.81	m/s^2
β	volume expansion coefficient ($1/T_f$ for gases)	1/K
Gr	Grashof number (buoyancy/viscous)	–
Ra	Rayleigh number Gr·Pr	–

Example: 0.6 m square plate, 60 K hotter than air: vertical $Ra = 7.65\times10^8$, $Nu = 0.59Ra^{1/4} = 98$, $h = 4.6$ W/m^{2}K, $Q = 99$ W.

Mass transfer (B7)

A species moves from high to low concentration by diffusion, exactly as heat moves from high to low temperature. **Fick's law** $J_A = -CD_{AB}dy_A/dx$ is Fourier's law with concentration for temperature and diffusivity for conductivity. Evaporation of a liquid into still air includes an extra bulk-flow effect (a log term).



Evaporation of a liquid through stagnant air

Symbol	Meaning	Unit
C	total molar concentration $P/(R_u T)$	kmol/m^3
y_A	mole fraction of A = P_A/P	–
D_{AB}	mass diffusivity of A in B	m^2/s
N_A, J_A	molar flux of A	kmol/m^{2}s
M	molar mass	kg/kmol
R_u	universal gas constant 8.314	kJ/kmol K
Sc, Sh	Schmidt ν/D_{AB}, Sherwood $h_m L/D_{AB}$	–

Example: air at 1 atm, 18.7 °C: $C = 101.325/(8.314 \times 291.85) = 0.0418$ kmol/m^3. Multiply a molar flux by M to get kg.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
BC	boundary condition
Bi	Biot number
CHF	critical heat flux
Fo	Fourier number
Gr	Grashof number
ID	inner diameter
kmol	kilomole
kPa	kilopascal
kW	kilowatt
LMTD	log mean temperature difference
Nu	Nusselt number
OD	outer diameter
Pr	Prandtl number
Ra	Rayleigh number
Re	Reynolds number
Sc	Schmidt number
Sh	Sherwood number
St	Stanton number